On the Invariant Theory of Forms in n Variables

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§ 1. Notation and Definitions. Summary of Known Results.

As usual, we call x, the row of elements $x_1, x_2, \dots, x_n$, a variable row. Analogously, p_ρ ($\rho = 1, 2, \dots, n-1$) denotes the variable row of the $\binom{n}{\rho}$ elements $p_{i_1 i_2 \dots i_\rho}$, where the $p_{i_1 i_2 \dots i_\rho}$ are the $\binom{n}{\rho}$ determinants

$$\begin{vmatrix} x_{i_1}^{(1)} & x_{i_2}^{(1)} & \cdots & x_{i_\rho}^{(1)} \\ x_{i_1}^{(2)} & x_{i_2}^{(2)} & \cdots & x_{i_\rho}^{(2)} \\ \vdots & \vdots & & \vdots \\ x_{i_1}^{(\rho)} & x_{i_2}^{(\rho)} & \cdots & x_{i_\rho}^{(\rho)} \end{vmatrix}, \qquad (i_1 < i_2 < \cdots < i_\rho; \quad i_1, i_2, \dots, i_\rho \in \{1, 2, \dots, n\}),$$

formed from the matrix of the ρ rows $x^{(1)}, x^{(2)}, \dots, x^{(\rho)}$. By the theorem on corresponding matrices,[1] to every row p_ρ there corresponds uniquely a second row $q_{n-\rho}$, through the relation valid for each element:

$$p_{i_1 i_2 \dots i_\rho} = (-1)^{\frac{\rho(\rho+1)}{2} + i_1 + i_2 + \cdots + i_\rho} q_{j_1 j_2 \dots j_{n-\rho}}, \tag{1}$$

where the i's and the j's are each ordered and together form a complete permutation of the numbers 1 through n. The row $q_{n-\rho}$ consists of the $\binom{n}{\rho}$ determinants of degree $n-\rho$, $q_{j_1 j_2 \dots j_{n-\rho}}$, formed from the matrix of $n-\rho$ rows $u^{(1)}, u^{(2)}, \dots, u^{(n-\rho)}$ contragredient to the x-rows. These rows satisfy the $\rho(n-\rho)$ conditions

$$(u^{(k)} \mid x^{(l)}) = \sum_{\alpha=1}^{n} u_\alpha^{(k)} x_\alpha^{(l)} = 0; \qquad (k = 1, 2, \dots, n-\rho;\ l = 1, 2, \dots, \rho). \tag{2}$$

Following Clebsch,[2] we assume them normalized so that the proportionality factor that would otherwise occur in (1) is equal to one.

According to Clebsch[3]—and likewise according to the later investigations of F. Deruyts and A. Capelli; cf. the Introduction—the most general forms can be reduced to forms containing at most one of each of the $n-1$ rows p_ρ. They can therefore be represented symbolically as

$$(S^1 \mid p_1)^{m_1} (S^2 \mid p_2)^{m_2} \cdots (S^{n-1} \mid p_{n-1})^{m_{n-1}}, \tag{3}$$

where, in analogy with (2), we have set

$$(S^\rho \mid p_\rho) = \sum S^{i_1 i_2 \dots i_\rho} p_{i_1 i_2 \dots i_\rho}.$$

Because of the nonlinear identities among the elements of a row p_ρ, the symbol rows S^ρ can be normalized so that they satisfy exactly the same identities. They then behave like the determinants of a matrix with ρ rows $s^{(1)}, s^{(2)}, \dots, s^{(\rho)}$ contragredient to the x-rows.[4] Thus to each row S^ρ there corresponds uniquely another row $S_{n-\rho}$ through the relation

$$S_{i_1 i_2 \dots i_{n-\rho}} = (-1)^{\frac{(n-\rho)(n-\rho+1)}{2} + i_1 + i_2 + \cdots + i_{n-\rho}} S^{j_1 j_2 \dots j_\rho}, \tag{4}$$

[1] Cf., for example, Gordan–Kerschensteiner, *Vorlesungen über Invariantentheorie*, vol. I, p. 99.
[2] Loc. cit., § 5.
[3] Loc. cit., § 12.
[4] Clebsch, loc. cit., § 4.

where again the i's and j's are each ordered and together exhaust the numbers 1 through n. The elements $S_{i_1 i_2 \ldots i_{n-\rho}}$ of the row $S_{n-\rho}$ behave like the determinants formed from $n-\rho$ rows $\sigma^{(1)}, \sigma^{(2)}, \ldots, \sigma^{(n-\rho)}$ cogredient to the x-rows, and the rows s and σ are linked by the $\rho(n-\rho)$ conditions

$$(s^{(k)} \mid \sigma^{(l)}) = 0, \qquad (k = 1, 2, \ldots, \rho;\ l = 1, 2, \ldots, n-\rho). \tag{5}$$

By (1) and (4), every factor of (3) admits the four equivalent symbolic representations

$$(S^\rho \mid p_\rho) = (S^\rho q_{n-\rho}) = (S_{n-\rho} p_\rho) = (-1)^{\rho(n-\rho)} (q_{n-\rho} \mid S_{n-\rho}),^{5} \tag{6}$$

Here and throughout what follows, $(S^\rho q_{n-\rho})$ and $(S_{n-\rho} p_\rho)$ abbreviate the determinants $(s^{(1)} \cdots s^{(\rho)} u^{(1)} \cdots u^{(n-\rho)})$ and $(\sigma^{(1)} \cdots \sigma^{(n-\rho)} x^{(1)} \cdots x^{(\rho)})$. Laplace expansion shows that these depend only on the rows $S^\rho, q_{n-\rho}$ and $S_{n-\rho}, p_\rho$, respectively, as the notation is intended to indicate. In accordance with the preceding discussion, $(S^\rho \mid p_\rho)$ and $(q_{n-\rho} \mid S_{n-\rho})$ denote the products of the matrices of the rows s and x, and of the rows u and σ, respectively. By a matrix product we mean the sum of the products of corresponding determinants, that is, the determinant of the product matrix.

Following Clebsch, we call the characteristic number $\rho(n-\rho)$ belonging to a symbol row (or variable row) its *weight*;[6] the number ρ, which gives the number of rows s, is its upper weight index, and the number $n-\rho$, which gives the number of rows σ, is its lower weight index. It follows from (4) that a symbol row can occur in the same relation once with an upper and once with a lower weight index; the same holds for the associated rows p_ρ and $q_{n-\rho}$. We further use the following general notation. Higher-order variable rows whose associated matrix consists of x-rows are denoted by p, and those whose matrix consists of u-rows by q. Symbol rows cogredient to the x-rows are denoted by lowercase Greek letters $\sigma, \tau, \alpha, \beta$; symbol rows cogredient to the u-rows by lowercase Latin letters s, t, a, b; and all other symbol rows by uppercase Latin letters carrying a weight index: S^ρ, T^τ, A^ρ, or $S_{n-\sigma}$, $T_{n-\tau}$, $A_{n-\rho}$. Correspondingly, the individual elements of a row are written with upper or lower indices according as the weight index is upper or lower, as in (4).

[5] Cf. Clebsch, loc. cit., § 5.
[6] Loc. cit., § 11.